

Project Report

Application of Fault Tree Analysis to Improving SWRO Desalination Plant Performance

By: Swathi Mohandas Surekha

1. Introduction

As water scarcity problems intensify every day due to population growth and the continuous groundwater depletion of freshwater quality, desalination plants were built around the world to provide potable water from seawater. The UAE, like other Middle Eastern countries, has met its growing water demand by desalinating Seawater using large-scale thermal distillation plants and the reverse osmosis process (RO). Thermal distillation requires a lot of power, so it must be run in conjunction with a power plant to save money. The demand for electricity in the winter is half that of the summer when air conditioning units across the country are running at total capacity. The gas must be burned in the summer to maintain the same output of desalinated water in the winter, driving up the cost of water to an unacceptably high price (Desalination, 2005).

Because of the rapid development in industries and population, the city of Fujarah in the United Arab Emirates has long had water supply issues. The Fujairah 1 Independent Water and Power Plant (F1 - IWPP) is a hybrid desalination plant that uses MSF and reverse osmosis (RO) technology. The Fujairah plant generates 12.5 MIGD water per day and has a 760 MW installed power capacity. The Fujairah facility utilizes a combination of two desalination processes to desalinate the water. Water is generated at a rate of 75.3 million gallons per day using multi-stage flash distillation (MSF) technology and 45.17 MIGD using reverse osmosis technology (RO) (ABB, 2005). The project also includes a potable water plant and five storage tanks, each with 20 MIG. Seawater desalination by reverse osmosis (SWRO) is becoming increasingly popular in the Middle East. Doosan Heavy Industries & Construction Company of South Korea served as the primary contractor in constructing Fujairah's water and power plant. In contrast, Degremont, a French business, planned, equipped, and oversaw the RO plant's assembly and initial operation (Desalination, 2005). Pretreatment removes mineral and organic micro-pollutants that could clog filtering membranes, increasing operating costs and decreasing membrane life.

Degremont created an innovative pretreatment system with three filter layers: pumice stone, sand, and different densities of sand (Bourouni, 2013). The Seawater Reverse Osmosis (SWRO) desalination plant comprises seawater intakes; a pretreatment module such as dual media, Sulphuric acid addition, cartridge filters; Reverse Osmosis a high-pressure feed pump, RO membranes rack such as seawater composite three Hydranautics membranes (First Train Pass) and Hydranautics Energy saving polyamide Membrane (Second Train Pass) and Brine Rejection; and lastly a post treatment module: treated tanks and pump.

The main goal of present reliability analysis is to ensure that the desired output level is maintained without any failures. System reliability management measures a system's performance and identifies the causes and frequency of failures. System failures or human errors can cause failures. Many continuously operating systems, such as the RO plant, can be repaired and maintained to a working condition. To restore system function and delay system failure, maintenance actions are required throughout the operation. The analyst must assess the component's reliability to understand the causes of accidents. There are three main approaches: failure modes and effects (FMEA), block diagram (E&F), and fault tree analysis (FTA).

This study uses Fault Tree Analysis (FTA) to evaluate the system's performance. Fault tree analysis is a logical model of an unwanted event's relationship to other events. This method creates a hierarchy with the undesirable event at the top, intermediate events in the middle, and essential causal events or primary events at the bottom. The most common logical symbols or gates are AND and OR. The AND gate requires all

succeeding events to occur before a preceding event, while the OR gate requires at least one succeeding event (Henley, E. J., & Kumamoto, 1992)

The increasing complexity of desalting plants necessitates design considerations for operation and maintenance. This report shows the application of fault tree techniques that can be used to improve desalination plant performance and water quality for the long term run of the plant (Sanza et al., 2007b). examined the two-year operational data from an RO plant in Fujairah and also general failure occur in each process in order to investigate the effect of the design and the operations conditions on the performance of the plant. An assessment is made of the impact of failures and outages on the availability of the RO desalination plant. The design configuration of the desalination plant and maintenance practices are found to impact the acceptability of water supply quality and reliability.

2. Methodology

The Fujairah 1 Seawater Reverse Osmosis Desalination plant is composed of the following system as shown in Figure 1:

a. Seawater Intakes:

The seawater intake is 380 m from the beach and 6 m above the seabed. One intake structure serves the RO plant while the other two serve the Multi Stage Flash Distillations and power plants, avoiding the continuous chlorination used on the other lines. The two pumps can feed 20,000 m³/h to the RO pretreatment. A cleanable screen and trash rake in the well prevent trash or fish from entering the two seawater supply pumps.

b. Pretreatment system:

The system consists of alum feed, dual media filters, Clearwell, sulfuric acid addition, phosphate addition, and cartridge filters. Two lines of seven open dual-media filters, each capable of handling up to 1,500 m³/h of Seawater and fed by gravity at 8.5 to 9.5 m/h. There are also two lines of nine cartridge filters, each having 360 cartridges of 5 microns. When required, an automatic system guarantees that doses of up to 25 mg/l H_2SO_4 , 5 mg/l $FeCl_3$, and 1.5 mg/l polymers are dosed proportionally to the incoming flow, but dose setpoints are manually adjusted concerning the observed seawater quality. Backwashing is triggered by time or pressure drop, and the filters are also operated automatically. The backwash process is a typical Degremont air scour followed by water rinsing at a high enough water flow rate to categorize the media by expanding both media both media therein. Such fine control of the expansion necessitates for the proper selection of the two media's materials (grain size and apparent density versus seawater); and capacity to adjust the backwash flow rate in response to changes in seawater viscosity.

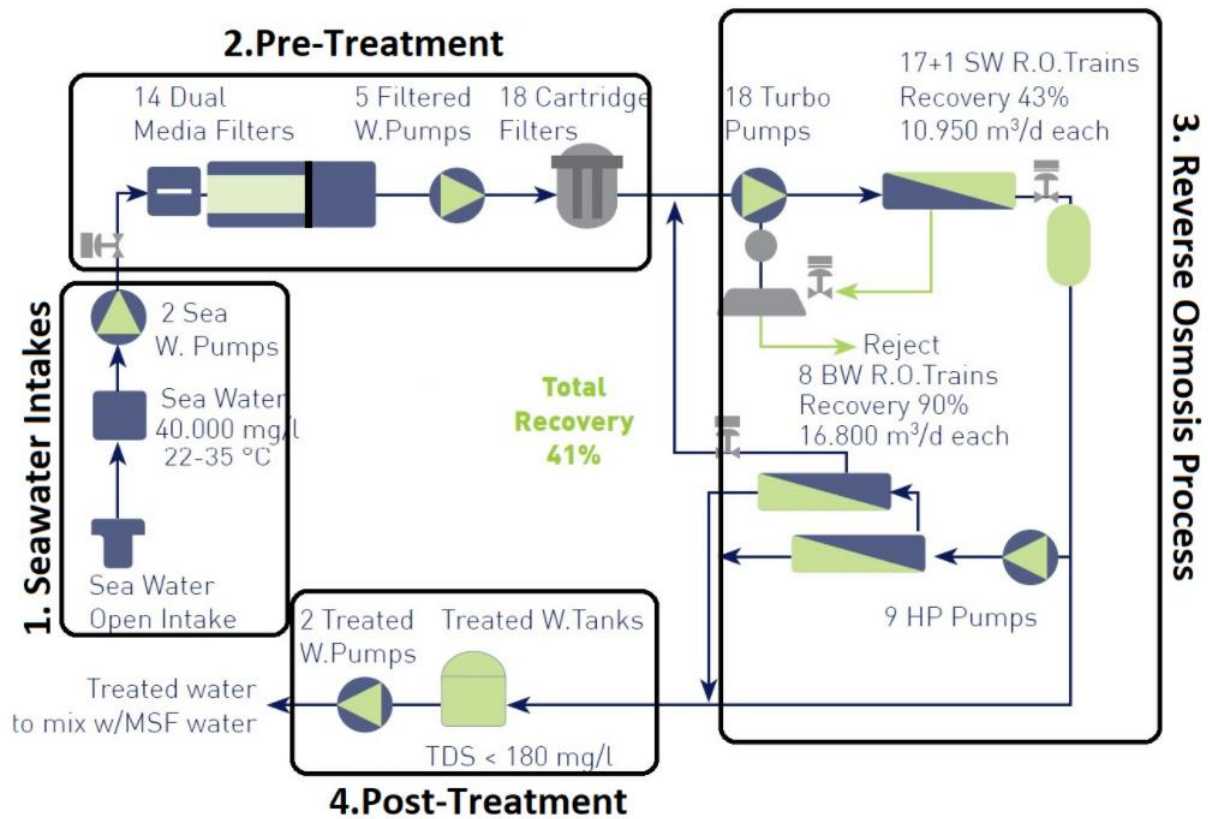


Figure 1: Block diagram of the four major systems of SWRO Plant of Fujairah I (Sanza et al., 2007b)

To modify the pH of the treated water, acid is injected into each stream. Injections of phosphate are used to prevent scale formation. Each stream is filtered after chemical treatment to improve chemical mixing and remove suspended particles. In two static mixers and two coagulation tanks, these chemicals are combined with saltwater.

c. Reverse osmosis system:

There are two stages to this system. The first stage is made up of nine units, and the second stage is made up of three. Each first-stage unit has two diesel engines and high-pressure pumps running in tandem, while each second-stage unit receives pressured feedwater from a single high-pressure feed pump. The Degrémont RO plant design features a double-pass system as follows: Because the required salinity at the SWRO plant output (before mixing with the MSF product) was less than 180 mg/L, the Degrémont RO plant incorporates a double-pass system as follows (Sanza et al., 2007b):

- First pass: 17 + 1 trains, each with 136 photovoltaics (PV) 7 elements Seawater Composite 3 (SWC3) Hydranautics membranes and a 43 percent recovery rate
- Second pass: 8 trains, each with 73 photovoltaics (PV) 7 elements Energy Saving Polyamide 1 (ESPA1) Hydranautics low energy membranes with a 90 percent recovery rate.

There are two stages to the trains (50 PV in the first stage and 23 in the second stage). About 20% of the 1st pass permeate flow is bypassed and blended with the 2nd pass permeate flow, resulting in blended water with a TDS of less than 180 mg/l. Because the brine produced by the second stage has a lower salinity than seawater (4–5 g/l vs. 38.5 g/l), it is recycled before the first pass, as indicated schematically in Figure 1. The first and second passes, as well as the pre-treatment, are designed in two separate lines.

d. Post-treatment system:

This system includes lime treatment, chlorine treatment, and a water distribution system. Lime is added to stabilize and raise the pH of the product water. Chlorine is added to sterilize the product water.

3. Result and Discussion

According to the flow diagram and process chain of the Seawater Reverse Osmosis desalination plant, the logical relation of the performance of different units to the top event occurrence (System Performance Failure) is analyzed. This analysis is based on experts and plant operators (Sanza et al., 2007a). Finally, the hierarchy of the factors leading to the occurrence of a top event is illustrated as the fault tree diagram. Figures 2 depict the fault tree diagram of the Seawater Reverse Osmosis Plant Process. According to Figure 3 explains the failure occurring in each Seawater Intakes, Pretreatment, Reverse Osmosis and Post treatment Process with the symbol for AND gate is “” and the symbol of OR gate is “”. These failures are categorized as human and mechanical errors, Operator error, Electrical Error, test or maintenance error, climate factor, In Human errors include operator absence and operator errors. Mechanical errors consist of equipment damages like (baffle, weir, and rotary bridge). Climate factors refer to atmospheric factors and high infiltration/inflow in the plant.

The fault tree diagram is divided into three figures (Figs. 3, 4, and 5) due to its large size and numerous components; a triangular symbol shows the connection of each figure to the next one. In these figures, the red color is for the top event, where yellow, gold, and dark blue represents intermediate events, and the grey color stands for basic events.

In the functioning of Seawater Intakes Process with no additional saltwater source, failure of this system is unacceptably dangerous. The seawater pump well contained foreign objects, sand, fish, and silt. Due to a buildup of debris in the pump well, the plant was forced to shut down several times due to low Seawater levels entering the well. Bearing and gland failure, suction issues or cavitation, various operation conditions, and maintenance conditions are the most common causes of pump failure. Corrosion, erosion, and cavitation are the most common failure modes for most pumps. Direct intake systems can provide substantial quantities of seawater supply if needed. Yet, overall experiences show that they are not as reliable a solution as it may seem. Massive biofouling is known to cause power plant shutdowns for safety reasons. Still, it appears that appropriate design changes can reduce the occurrence of such events at the expense of reduced power plant efficiency. (Shahabi et al., 2015)

The filter, pump, or motor failure in the Pretreatment process and an inappropriate dosage agent causes the incapacity to function for an extended period. However, most filter issues, such as corrosion, pore occlusion, and surface reactivity, can result in high maintenance costs, process failures, or product contamination. Corrosive fluids can corrode the filter element, causing damage and eventual failure. Corrosion will damage the filter's function and taint the stream that passes through it. Metal ion pollution

from a metallic filter is a severe hazard to applications, regardless of whether or not there is measurable corrosion. Chemical adsorption and catalytic activity would damage the process stream if the media passing through the filter reacted with the filter substrate. The filter element's chemical adsorption might cause product quality issues and analytical test failures. Finally, as media accumulates within the filter, any filtration media can eventually suffer from pore blockage and flow limitation. As a result, filtration efficiency would deteriorate with time, necessitating filter replacement.

The most common causes of reverse osmosis issues include pipe leakage, pumps, diesel engines, and instrumentation failure. Several factors contributed to these failures, including improper material selection, poor operating conditions, wrong design, and poor maintenance. Lack of maintenance and defective replacement parts, which take a long time to order, also contribute to plant outages. Bearing failure, mechanical seal failure, defective oil seal failure, and leaks are the most common causes of pump failure. And also, the head of the high-pressure pump selected in the initial design is low, which causes the water production to fail to meet the design requirements when the temperature or the quality of the feed water changes.

Membranes deteriorate and must be replaced as undesired materials accumulate on their surface. Degradation through chemical (oxidation, hydrolysis, etc.) and mechanical (compaction, telescoping) activities may also contribute to a decline in membrane performance. A significant success factor for Reverse Osmosis (RO) systems is extending membrane life and lowering replacement costs. The most common cause of RO system failure is fouling caused by foulant deposition on the membrane surface. The deposition of numerous separate but related foulants on the membrane surface is referred to as RO membrane fouling. Fouling issues in RO systems are becoming more common as the usage of low-quality input water grows. Furthermore, water treated with various organic compounds causes substantial and difficult fouling issues. When the performance of a RO system deteriorates, the operating costs rise. Membrane cleanup, replacement, and system downtime are all included in these expenditures. System design, pretreatment (e.g., chemical conditioning), and system maintenance are all essential variables in the effectiveness of a RO system. Other failures include Damage to the sealing ring causes a decrease in the quality of the produced water; the mixed-use of new and old membrane elements and different types of membrane elements cause system performance degradation; the design of reverse osmosis recovery rate is unreasonable, and the number of membrane elements is too small; System shutdown causes pollutant deposition and bacterial and microbial pollution.

The treated water does not meet the standard water quality requirement due to improper control of pH, corrosion and dosing chlorine or ozone gas in the post treated process. If conductive of permeate water or treated water is high, it causes membrane defective, mechanical, and valve failure. When permeate water is needed for drinking or industrial reasons, mineral dosing or blending is recommended. Both operations are suggested based on the quality of the raw water. Blending is recommended only if the water quality is acceptable enough to drink; there should be no microorganisms in the water that is mixed with permeate water in a certain ratio to achieve the appropriate TDS level. Mineral dosage is recommended if the water quality is poor. A study comparing the scaling tendency of raw water and synthetic water due to salts found that synthetic water has more strength to form scales at higher temperatures. After all of the steps have been performed, disinfection must eliminate all bacteria from the water. Chemical dosing or UV lamp disinfection are both options for disinfection. Chlorine or ozone gas dosage are examples of chemical disinfection. In an ultraviolet lamp, germs are killed by passing treated water over the lamp. Membrane rejects water can be kept and used for domestic and industrial uses. If zero discharge is desired, open pits

with water evaporated and salts left behind are preferred. Because the concentration of reject water is modified by mixing, using rejection with feed is efficient. A TDS limit has been set, and when this limit is exceeded, a comparable amount of reject water is added. Rejection recovery plans play a significant role in reverse osmosis costs (Idrees, 2020).



Figure 2 : Seawater Reverse Osmosis System Performance Failure in each process

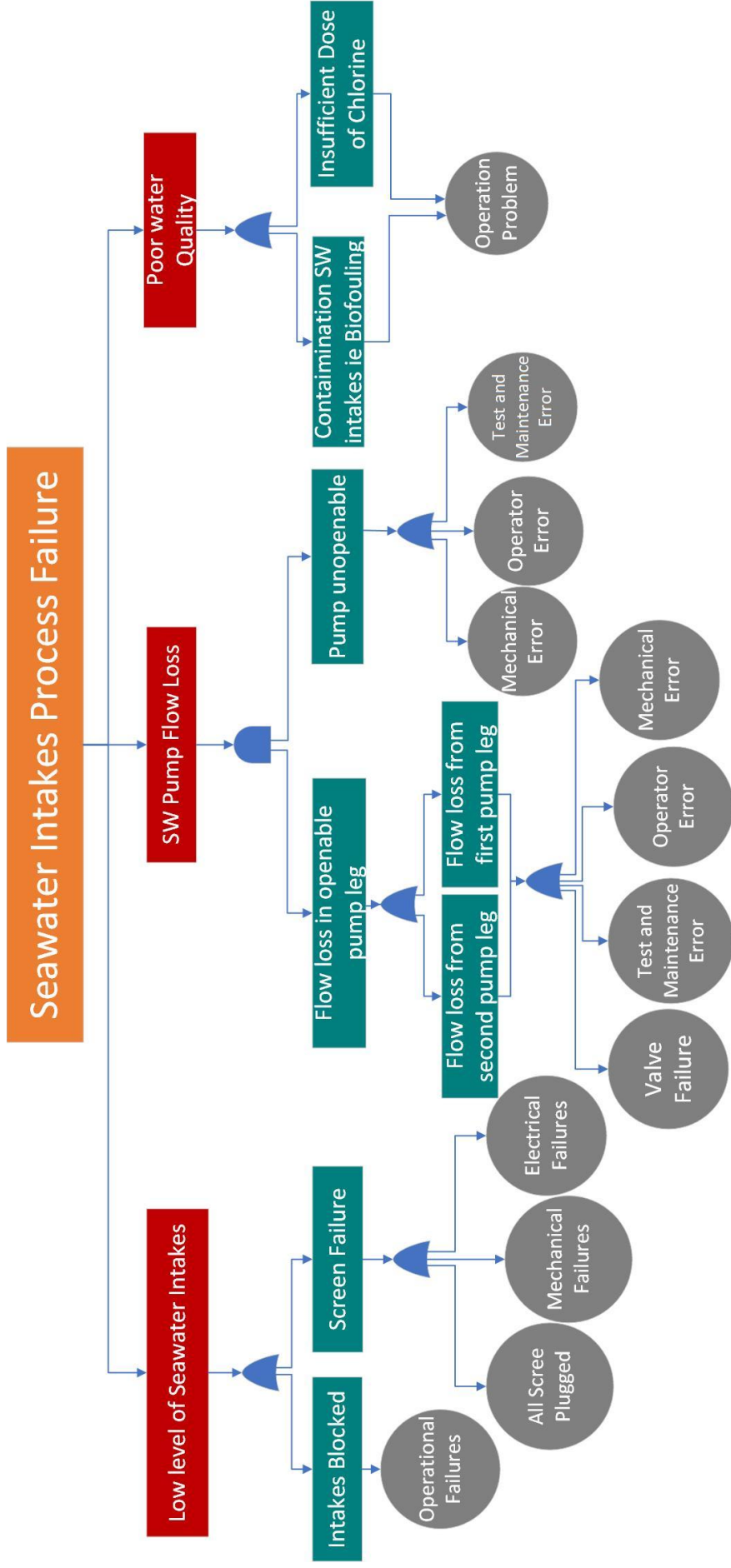


Figure 3 : Failure Occurred in Seawater Intakes

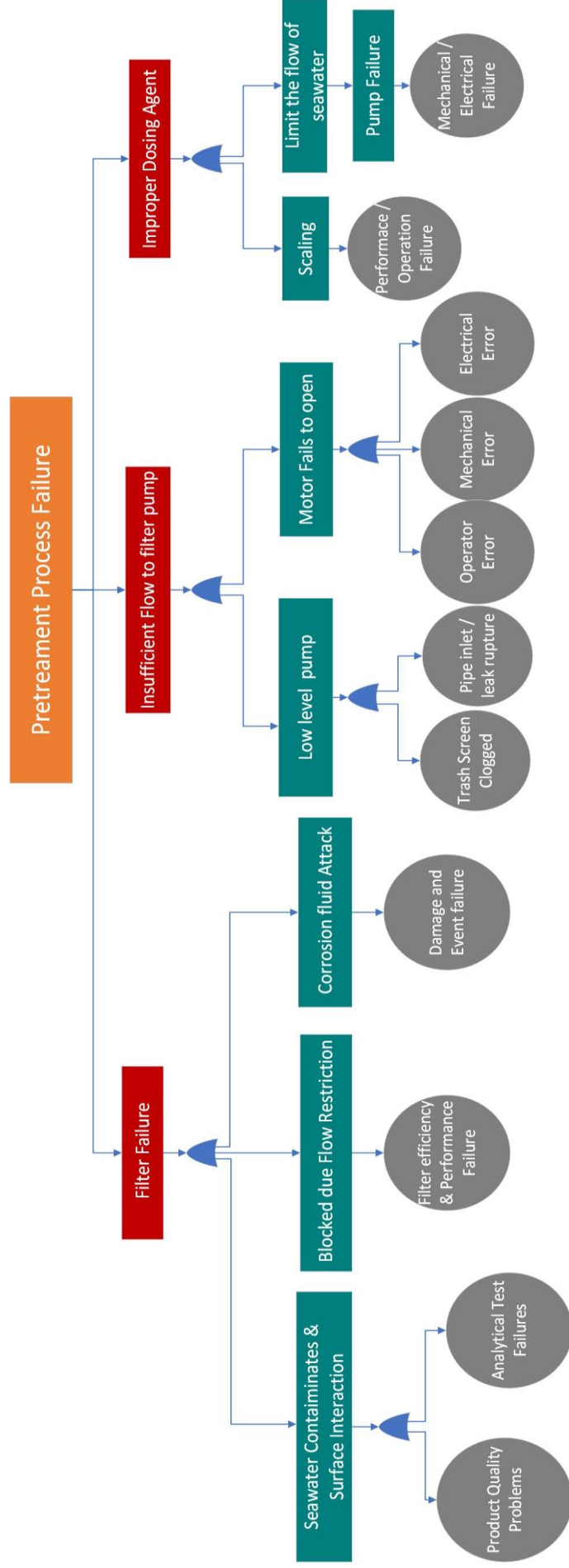


Figure 4: Failure Occurred in Pretreatment Process

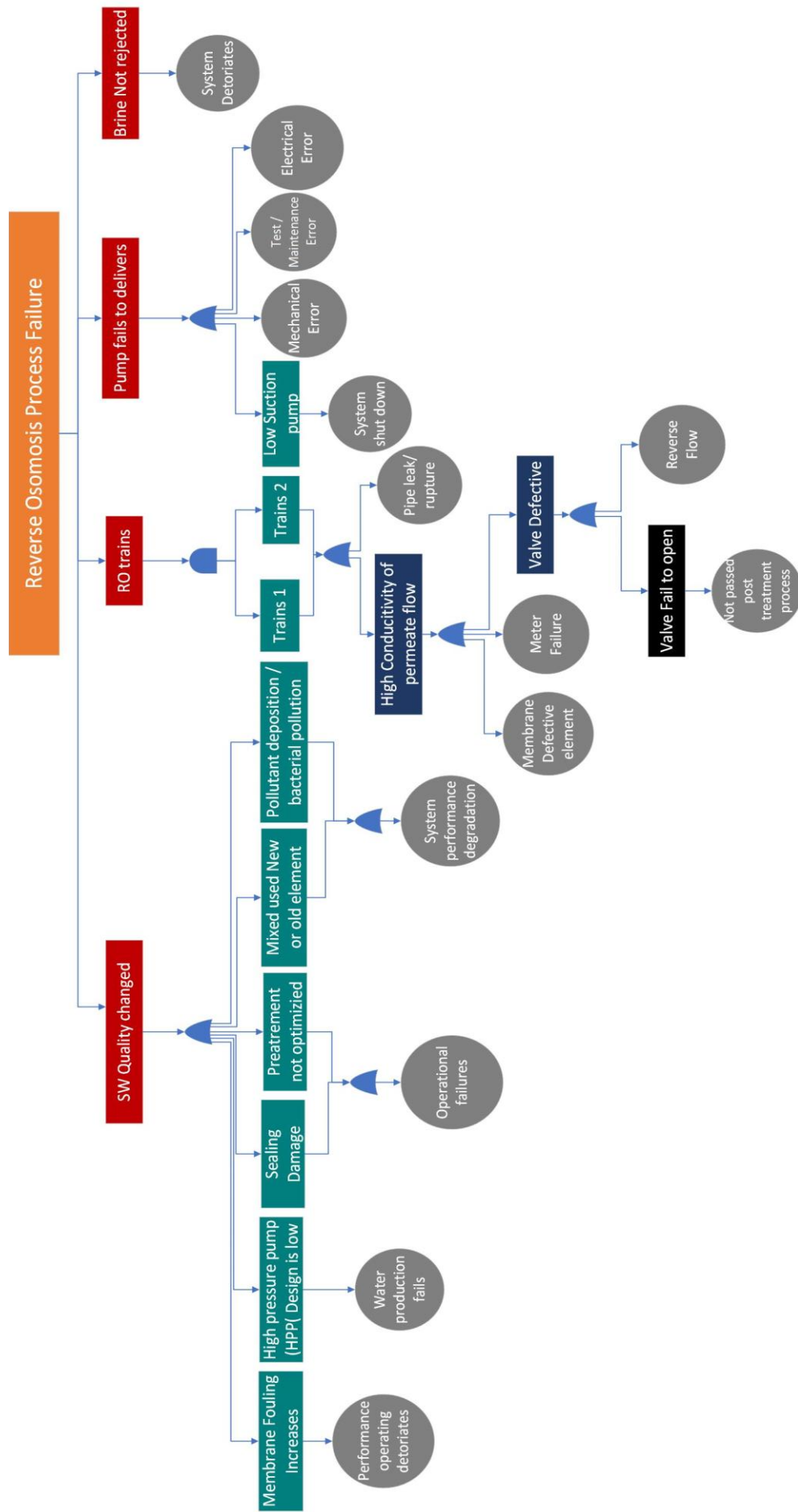


Figure 5 : Failure Occurred in Reverse Osmosis Process

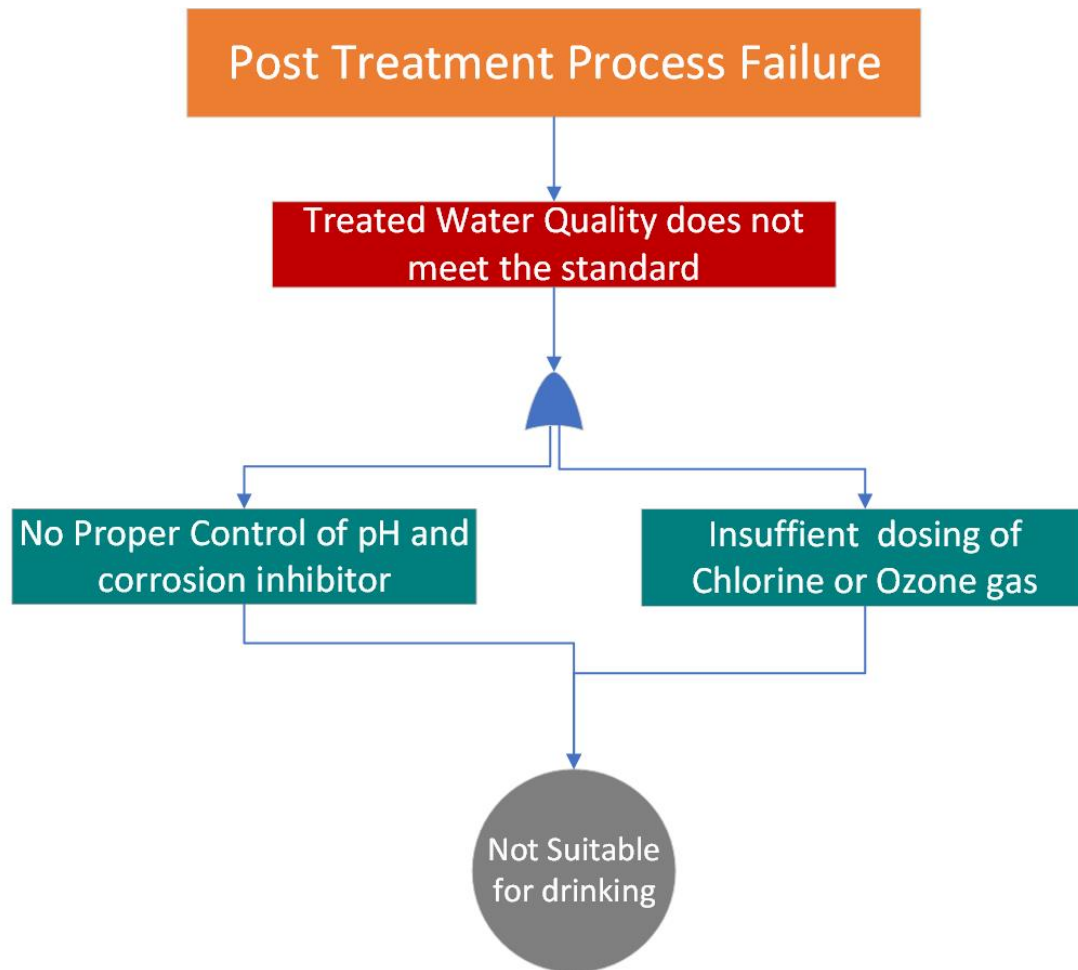


Figure 6 : Failure Occurred in Post-Treatment Process

4. Conclusion and Recommendation:

Seawater reverse Osmosis (SWRO) has become the most extensively used method in the desalination plants worldwide because it is least expensive than the Distillation process. It is especially well suited to countries that face water scarcity and have limited energy supplies. Desalination plants are becoming more sophisticated, necessitating design concerns for operation and maintenance. This study demonstrates how fault tree approaches can improve desalination plant performance and water quality to ensure the plant's long-term viability. (Sanza et al., 2007b) looked analyzed two years of operational data from a RO plant in Fujairah. The design and operating conditions affected the plant's performance and produced a Fault Tree Analysis (FTA) analysis report. The fault tree analysis of these systems provided that high-pressure pumps and diesel engines failures, pipes leakage, and membrane defects are the plant's dominant contributors. The full-scale plant results demonstrate high performance and achievement of excellent water quality. Frequent filter membrane element replacement has several advantages for plant operations, including higher recovery and train capacity and improved product water quality. These facilities also require a good operation team with knowledge and the ability to execute a practical feedback of the seawater, pre-

treatment, and membrane conditions in real time to optimize all operational parameters and operating costs for the next years.

5. Reference

- ABB. (2005). *Performance Monitoring & Optimization*.
- Bourouni, K. (2013). Availability assessment of a reverse osmosis plant: Comparison between Reliability Block Diagram and Fault Tree Analysis Methods. *Desalination*, 313, 66–76.
<https://doi.org/10.1016/j.desal.2012.11.025>
- Desalination. (2005, February 5). Fujairah marks major milestone for desalination in Middle East. . *WaterWorld*.
- Henley, E. J., & Kumamoto, H. (1992). Probabilistic risk assessment: Reliability engineering, design, and analysis. *IEEE*.
- Idrees, M. F. (2020). Performance Analysis and Treatment Technologies of Reverse Osmosis Plant – A case study. *Case Studies in Chemical and Environmental Engineering*, 2, 100007.
<https://doi.org/10.1016/J.CSCEE.2020.100007>
- Sanza, M. A., Bonnélyea, V., & Cremerb, G. (2007a). Fujairah reverse osmosis plant: 2 years of operation. *Desalination*, 203(1–3), 91–99. <https://doi.org/10.1016/j.desal.2006.03.526>
- Sanza, M. A., Bonnélyea, V., & Cremerb, G. (2007b). Fujairah reverse osmosis plant: 2 years of operation. *Desalination*, 203(1–3), 91–99. <https://doi.org/10.1016/j.desal.2006.03.526>
- Shahabi, M. P., McHugh, A., & Ho, G. (2015). Environmental and economic assessment of beach well intake versus open intake for seawater reverse osmosis desalination. *Desalination*, 357, 259–266.
<https://doi.org/10.1016/J.DESAL.2014.12.003>